

SHOULD-COST ANALYSIS REPORT

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Bottom-Up Manufacturing Cost Model · Fully Traceable Rate Data

steering_knuckle_RH

Commodity: CAST AND MACHINE · Currency: CNY · FX Rate: 9.0498 to GBP · Operations: 6 · Region: CN

Total Should-Cost	Material	Process	Labour	Margin
¥206.21	19.2%	36.3%	16.0%	7.4%

Model Confidence: **Low** · 6 traced operations · 15 data points auditable

Engine Warnings (1)

rawMaterial.materialId: Material rate confidence: Low

Geometry Provenance — MEASURED (OCCT B-rep kernel)

Volume, weight and every feature were measured from the CAD solid by the Open CASCADE kernel. Measured volume 356.1 cm³; mass for the selected material family 2.528 kg.

Material cost and geometry-derived machining are grounded in the actual solid — not an estimate.

Key Assumptions

Annual volume: 200,000 · Region: CN · Commodity: cast and machine

Alloy / material: EN-GJS-500-7 (Ductile Iron) · Net weight: 2.625 kg (measured)

Material utilisation: 65% · Overhead: 12% of factory base · Margin: 8% of subtotal · Operations: 6

§1 — 8-Bucket Cost Breakdown

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
1. Raw Material	¥39.54	19.2%	
2. Process (Machine)	¥74.95	36.3%	
3. Direct Labour	¥32.97	16.0%	
4. Tooling (amortised)	¥19.55	9.5%	
5. Packaging	¥1.45	0.7%	

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
6. Logistics	¥2.44	1.2%	
Factory Cost	¥170.90	82.9%	
7. Overhead (SG&A)	¥20.04	9.7%	
Subtotal	¥190.94	92.6%	
8. Supplier Margin	¥15.28	7.4%	
TOTAL SHOULD-COST	¥206.21	100.0%	

§2 — Commercial Parameters

Overhead Rate	12.0%	Supplier Margin	8.0%
Packaging / part	¥1.45	Logistics / part	¥2.44
Total Tooling Cost	¥3909502.26	Amortisation Volume	200,000 parts

§3 — Material Detail

Parameter	Value	Unit	Notes
Material ID	mat-gjs500	ID	
Grade / Specification	EN-GJS-500-7 (Ductile Iron)		Index-anchored 2026-07 (see CASTING PRICING BASIS). Mid-grade ductile iron — steering knuckles, hubs, brackets; balance of strength and ductility. Regional adj. ×0.880 (millSteel)
Region	China		
Net Finished Weight	2.6247 kg	kg	Weight in finished part
Gross Weight (stock)	4.0381 kg	kg	Net ÷ utilisation ratio
Scrap / Runner Weight	1.4133 kg	kg	Gross - Net
Material Utilisation	65.0%	%	Benchmark: casting 65–85 % (runners + risers)
Material Price	¥6.85	CNY /kg	Index-anchored 2026-07 (see CASTING PRICING BASIS). Mid-grade ductile iron — steering knuckles, hubs, brackets; balance of strength and ductility. Regional adj. ×0.880 (millSteel)
Scrap Recovery Price	¥1.19	CNY /kg	
Gross Material Cost	¥27.66	CNY	Gross × price/kg
Scrap Credit	-¥1.69	CNY	Scrap × recovery price
Consumables (core / wax / shell)	¥13.57	CNY	Per-part recurring
NET RAW MATERIAL COST	¥39.54	CNY	Gross - scrap credit + consumables

Data Confidence	Low
Effective Date	2026-07

§4 — Operations Detail

4A Machine Operations

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
Sand Casting — Moulding	Sand Casting Moulding Line	¥136.44	11.66	1	80.0%	¥33.14
Machining Setup (amortised)	HAAS VF-2 (3-axis Mill)	¥212.31	0.90	1	100.0 %	¥3.18
Milling — +Z (60 faces)	HAAS VF-2 (3-axis Mill)	¥212.31	1.10	1	85.0%	¥4.60

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
Milling — +X (51 faces)	HAAS VF-2 (3-axis Mill)	¥212.31	0.94	1	85.0%	¥3.90
Milling — +Y (30 faces)	HAAS VF-2 (3-axis Mill)	¥212.31	0.55	1	85.0%	¥2.30
Drilling — 14 holes (1×Ø10.0×12, 1×Ø11.0×5, 1×Ø11.0×16, 1×Ø12.0×2, 2×Ø12.5×16, 1×Ø12.5×25, 1×Ø15.0×19, 1×Ø15.0×25, 1×Ø63.0×3, 1×Ø68.0×4, 1×Ø68.0×37, 1×Ø71.0×3, 1×Ø75.0×6) [geometry-measured]	CNC Drilling Centre	¥136.82	10.37	1	85.0%	¥27.83
TOTAL						¥74.95

4B Labour Detail

Operation	Labour Grade	Rate / hr	Man ning	Lab. min	Eff %	Labour Cost	Op Total
Sand Casting — Moulding	Skilled Machinist	¥71.49	1	11.66	92.0 %	¥15.10	¥48.25
Machining Setup (amortised)	Skilled Machinist	¥71.49	1	0.90	100.0 %	¥1.07	¥4.26
Milling — +Z (60 faces)	Skilled Machinist	¥71.49	1	1.10	92.0 %	¥1.43	¥6.03
Milling — +X (51 faces)	Skilled Machinist	¥71.49	1	0.94	92.0 %	¥1.21	¥5.11
Milling — +Y (30 faces)	Skilled Machinist	¥71.49	1	0.55	92.0 %	¥0.71	¥3.01
Drilling — 14 holes (1×Ø10.0×12, 1×Ø11.0×5, 1×Ø11.0×16, 1×Ø12.0×2, 2×Ø12.5×16, 1×Ø12.5×25, 1×Ø15.0×19, 1×Ø15.0×25, 1×Ø63.0×3, 1×Ø68.0×4, 1×Ø68.0×37, 1×Ø71.0×3, 1×Ø75.0×6) [geometry-measured]	Skilled Machinist	¥71.49	1	10.37	92.0 %	¥13.44	¥41.27
TOTAL						¥32.97	¥107.92

4C Machined Features — Geometry Audit

Feature	Ø×depth / area	Qty	Operation	Min	Cost	Costed ?
Boss	Ø25.0 × 17	2	Turning (external Ø)	1.28	—	No
Boss	Ø31.0 × 3	1	Turning (external Ø)	0.36	—	No
Boss	Ø31.0 × 26	1	Turning (external Ø)	0.82	—	No
Boss	Ø78.0 × 16	1	Turning (external Ø)	0.62	—	No
Hole / bore	Ø10.0 × 12 blind	1	Drilling (blind)	0.48	¥2.05	Yes
Hole / bore	Ø11.0 × 5 blind	1	Drilling (blind)	0.35	¥1.48	Yes

Feature	Ø×depth / area	Qty	Operation	Min	Cost	Costed ?
Hole / bore	Ø11.0 × 16 blind	1	Drilling (blind)	0.56	¥2.37	Yes
Hole / bore	Ø12.0 × 2 blind	1	Drilling (blind)	0.29	¥1.23	Yes
Hole / bore	Ø12.5 × 16 blind	2	Drilling (blind)	1.14	¥4.83	Yes
Hole / bore	Ø12.5 × 25 blind	1	Drilling (blind)	0.75	¥3.18	Yes
Hole / bore	Ø15.0 × 19 blind	1	Drill + bore (blind)	1.26	¥5.32	Yes
Hole / bore	Ø15.0 × 25 blind	1	Drill + bore (blind)	1.50	¥6.36	Yes
Hole / bore	Ø63.0 × 3 blind	1	Drill + bore (blind)	0.45	¥1.92	Yes
Hole / bore	Ø68.0 × 4 blind	1	Drill + bore (blind)	0.47	¥2.00	Yes
Hole / bore	Ø68.0 × 37 blind	1	Drill + bore (blind)	2.42	¥10.28	Yes
Hole / bore	Ø71.0 × 3 blind	1	Drill + bore (blind)	0.48	¥2.05	Yes
Hole / bore	Ø75.0 × 6 blind	1	Drill + bore (blind)	0.50	¥2.12	Yes
Face	1544 mm²	1	Face milling (facing)	0.39	¥1.67	Yes
Face	1298 mm²	1	Face milling (facing)	0.36	¥1.53	Yes
Face	1184 mm²	1	Face milling (facing)	0.35	¥1.47	Yes
Face	1161 mm²	1	Face milling (facing)	0.34	¥1.46	Yes
Face	1012 mm²	1	Face milling (facing)	0.33	¥1.39	Yes
Face	1001 mm²	1	Face milling (facing)	0.33	¥1.38	Yes
Face	962 mm²	1	Face milling (facing)	0.32	¥1.36	Yes
Face	956 mm²	1	Face milling (facing)	0.32	¥1.36	Yes
Pocket	1544 mm² × 20	1	Pocket milling (rough + finish)	5.71	—	No
Pocket	1161 mm² × 34	1	Pocket milling (rough + finish)	7.26	—	No
Pocket	913 mm² × 34	1	Pocket milling (rough + finish)	5.82	—	No
Pocket	758 mm² × 7	1	Pocket milling (rough + finish)	1.18	—	No
Pocket	491 mm² × 68	1	Pocket milling (rough + finish)	6.37	—	No
TOTAL costed		22		13.40	¥56.81	Yes

10 detected feature(s) NOT costed

10 machinable feature instance(s) (29.4 min) were detected in the geometry but are NOT included in this cost — on a near-net part they may be as-cast. If they are machined, tick them in the Machined-Features panel and re-cost; this could add up to ¥124.69.

\$5 — Machine Rate Buildup

CNC Drilling Centre [mach-drill] · Confidence: Low · UK Tier-2 CNC drilling/tapping centre. Target £30/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥139366.52	¥43.55	4,000 hr/yr available
Maintenance	¥59728.51	¥18.67	
Energy	¥49577.02	¥15.49	
Floor Space	¥39819.00	¥12.44	
Indirect Support	¥79638.01	¥24.89	
Finance Cost	¥69683.26	¥21.78	
TOTAL — CNC Drilling Centre	¥437812.32	¥136.82	Effective hours: 3200/yr

HAAS VF-2 (3-axis Mill) [mach-haas-vf2] · Confidence: Low · HAAS VF-2 benchmark, UK Tier-2 machining shop | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @50% util	Notes
Depreciation	¥189140.27	¥94.57	4,000 hr/yr available
Maintenance	¥89592.76	¥44.80	
Energy	¥28919.93	¥14.46	
Floor Space	¥24886.88	¥12.44	
Indirect Support	¥47285.07	¥23.64	
Finance Cost	¥44796.38	¥22.40	
TOTAL — HAAS VF-2 (3-axis Mill)	¥424621.29	¥212.31	Effective hours: 2000/yr

Sand Casting Moulding Line [sand-cast-line] · Confidence: Low · UK foundry benchmark, Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @75% util	Notes
Depreciation	¥124434.39	¥47.40	3,500 hr/yr available
Maintenance	¥59728.51	¥22.75	
Energy	¥68856.97	¥26.23	
Floor Space	¥49773.76	¥18.96	
Indirect Support	¥39819.00	¥15.17	
Finance Cost	¥15554.30	¥5.93	
TOTAL — Sand Casting Moulding Line	¥358166.93	¥136.44	Effective hours: 2625/yr

How to read the machine rate

Rate/hr = (annual depreciation + maintenance + energy + floor + indirect + finance) ÷ effective hours, where effective hours = available hours × utilisation.

Effective hours encode the shift pattern: a lower hours-base raises £/hr and a higher one lowers it. A challenge from a supplier will target this divisor and whether depreciation is machine-only or full-cell (machine + dosing furnace + trim + robots/extraction) — state your basis when defending the number.

§6 — Rate Traceability

Field	Value	Unit	Source / Reference	Rate ID	Conf.
material.pricePerKg	0.7568	£/kg	Index-anchored 2026-07 (see CASTING PRICING BASIS). Mid-grade ductile iron — steering knuckles, hubs, brackets; balance of strength and ductility. Regional adj. ×0.880 (millSteel)	mat-gjs500	Low
material.scrapRecoveryPricePerKg	0.1320	£/kg	Index-anchored 2026-07 (see CASTING PRICING BASIS). Mid-grade ductile iron — steering knuckles, hubs, brackets; balance of strength and ductility. Regional adj. ×0.880 (millSteel)	mat-gjs500	Low
rawMaterial.consumablesCostPerPart	1.5000	£	Per-part consumable (core/wax/shell)	mat-gjs500	Medium
Sand Casting — Moulding.machineRatePerHr	15.0771	£/hr	UK foundry benchmark, Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	sand-cast-line	Low
Sand Casting — Moulding.labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Machining Setup (amortised).machineRatePerHr	23.4603	£/hr	HAAS VF-2 benchmark, UK Tier-2 machining shop Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-haas-vf2	Low
Machining Setup (amortised).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Milling — +Z (60 faces).machineRatePerHr	23.4603	£/hr	HAAS VF-2 benchmark, UK Tier-2 machining shop Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-haas-vf2	Low
Milling — +Z (60 faces).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Milling — +X (51 faces).machineRatePerHr	23.4603	£/hr	HAAS VF-2 benchmark, UK Tier-2 machining shop Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-haas-vf2	Low
Milling — +X (51 faces).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Milling — +Y (30 faces).machineRatePerHr	23.4603	£/hr	HAAS VF-2 benchmark, UK Tier-2 machining shop Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-haas-vf2	Low
Milling — +Y (30 faces).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Drilling — 14 holes (1×Ø10.0×12, 1×Ø11.0×5, 1×Ø11.0×16, 1×Ø12.0×2, 2×Ø12.5×16, 1×Ø12.5×25, 1×Ø15.0×19, 1×Ø15.0×25, 1×Ø63.0×3, 1×Ø68.0×4, 1×Ø68.0×37, 1×Ø71.0×3, 1×Ø75.0×6) [geometr	15.1182	£/hr	UK Tier-2 CNC drilling/tapping centre. Target £30/hr. Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-drill	Low

Field	Value	Unit	Source / Reference	Rate ID	Conf
y-measured].machineRatePerHr					
Drilling — 14 holes (1×Ø10.0×12, 1×Ø11.0×5, 1×Ø11.0×16, 1×Ø12.0×2, 2×Ø12.5×16, 1×Ø12.5×25, 1×Ø15.0×19, 1×Ø15.0×25, 1×Ø63.0×3, 1×Ø68.0×4, 1×Ø68.0×37, 1×Ø71.0×3, 1×Ø75.0×6) [geometr y-measured].labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low

Estimate (± band)	P10 · optimistic	P50 · median	P90 · conservative	Band	CV
¥206.21 ± 16.2%	¥173.94	¥204.16	¥241.00	moderate	12.7%

Range reflects how well each cost driver is known (High / Medium / Low confidence). Tighten it by attaching CAD, a BOM, or logging an actual quote.

Why confidence is Low

Main band drivers: 14 rate(s) at Low confidence; tooling amortisation carries the widest per-bucket spread; some detected features not costed (see §4C).

What's excluded from this unit cost

One-time NRE — tooling / die, fixtures and CNC programming — is amortised separately, not in this unit cost.

Import duty and international freight (regional table is Ex-Works).

Formal embodied-carbon reporting (figures are indicative cradle-to-gate — replace with supplier EPDs).

Heat-treat (T6/T7), HIP and 100% NDT/X-ray are NOT in this cost — add them for structural / safety-critical parts.

Trim/degate, shot-blast/deflash and impregnation are only included where shown as operations.

§8 — Sensitivity Analysis

± 10% driver swing · ranked by ¥ impact

Cost Driver	Base	-10% cost	+10% cost	Range ¥
Sand Casting — Moulding: Cycle Time	0.19hr	-2.8%	+2.8%	¥11.67
Drilling — 14 holes (1×Ø10.0×12, 1×Ø11.0×5, 1×Ø11.0×16, 1×Ø12.0×2, 2×Ø12.5×16, 1×Ø12.5×25, 1×Ø15.0×19, 1×Ø15.0×25, 1×Ø63.0×3, 1×Ø68.0×4, 1×Ø68.0×37, 1×Ø71.0×3, 1×Ø75.0×6) [geometry-measured]: Cycle Time	0.17hr	-2.4%	+2.4%	¥9.98
Sand Casting — Moulding: Machine Rate	¥136.44/hr	-1.9%	+1.9%	¥8.02
Sand Casting — Moulding: Labour Rate	¥71.49/hr	-1.9%	+1.9%	¥7.98
Machining Setup (amortised): Labour Rate	¥71.49/hr	-1.9%	+1.9%	¥7.98
Milling — +Z (60 faces): Labour Rate	¥71.49/hr	-1.9%	+1.9%	¥7.98
Milling — +X (51 faces): Labour Rate	¥71.49/hr	-1.9%	+1.9%	¥7.98
Milling — +Y (30 faces): Labour Rate	¥71.49/hr	-1.9%	+1.9%	¥7.98

§9 — Regional Cost Comparison

Ex-Works · per-region should-cost · vs China

Region	Material	Process	Labour	Tooling	Overhead	Ex-Works	Logistics	Total	vs China
United Kingdom (GBP)	¥44.93	¥136.27	¥118.68	¥35.54	¥26.72	¥362.15	¥1.69	¥381.18	+85%
Germany (EUR)	¥46.28	¥143.08	¥191.81	¥37.32	¥36.67	¥455.16	¥1.94	¥478.45	+132%
France (EUR)	¥45.83	¥125.37	¥137.86	¥32.70	¥28.59	¥370.35	¥1.94	¥390.09	+89%
Spain (EUR)	¥44.93	¥119.92	¥86.91	¥31.28	¥21.42	¥304.46	¥1.94	¥321.39	+56%
Poland (PLN)	¥43.59	¥98.12	¥53.95	¥25.59	¥14.98	¥236.22	¥2.02	¥250.13	+21%
Turkey (TRY)	¥40.44	¥81.76	¥29.97	¥21.32	¥10.78	¥184.28	¥2.19	¥195.98	-5%
China (CNY)	¥39.54	¥74.95	¥32.97	¥19.55	¥20.04	¥187.05	¥2.44	¥206.21	Base
India (INR)	¥40.44	¥70.86	¥20.98	¥18.48	¥8.65	¥159.41	¥2.53	¥170.10	-18%
Mexico (MXN)	¥42.69	¥81.76	¥34.77	¥21.32	¥11.22	¥191.76	¥2.27	¥203.75	-1%
United States (USD)	¥44.93	¥115.83	¥155.84	¥30.21	¥26.25	¥373.07	¥2.11	¥392.88	+91%

Per-region should-cost using regional labour, energy and rate benchmarks. Tooling held fixed. Ex-Works — excludes import duty + international freight. Indicative — confirm with RFQ.

§10 — Embodied Carbon

17.93 kgCO2e · 6.83 /kg · cradle-to-gate

Material	5.96 kgCO2e	Material factor	2.10 kg/kg (Steel)
Process	11.55 kgCO2e	Grid intensity	0.550 kg/kWh · 21.00 kWh
Logistics	0.42 kgCO2e	Total	17.93 kgCO2e

Indicative cradle-to-gate factors — replace with supplier EPDs for formal reporting.

[WARNING] Machining / process cost above benchmark

Finding	Process cost at 36.3% of total is above the benchmark range of 8–18%. Cycle time or machine rate is elevated.
Impact	High · up to 9% potential saving
Benchmark	Process %: yours 36.3% vs industry 8–18%
Action 1	Reduce cycle time via feeds/speeds optimisation and toolpath strategy review
Action 2	Combine multiple operations into one setup to eliminate re-fixturing time

[WARNING] Elevated process cost — possible high reject/scrap rate

Finding	Process cost is 36.3% vs benchmark max of 18%. For castings, this may indicate a high reject rate (>5%) inflating effective machine time, poor OEE on the casting cell, or cycle time above industry benchmark for this alloy/weight range.
Impact	Medium · up to 7% potential saving
Action 1	Audit actual first-pass yield (FPY) data from the foundry — target >96% for HPDC Al
Action 2	Root-cause the dominant reject type: porosity, dimensional, surface, or cold shut

[WARNING] Low material utilisation — high scrap rate

Finding	Material utilisation at 65% is below the cast and machine benchmark of 82%. You are paying for 54% more raw material than ends up in the part.
Impact	Medium · up to 3% potential saving
Action 1	Redesign blank/preform geometry to minimise offcuts
Action 2	Optimise nesting/layout for sheet or profile stock

[WARNING] Low casting yield — excess runner/gating material

Finding	Casting yield of 65% means 54% of poured metal is returned as runner/gate scrap. HPDC benchmark is 65–75%; sand/gravity 75–85%. Scrap recovery credits are partial — you are paying for metal that doesn't end up in the part.
Impact	Medium · up to 2% potential saving
Benchmark	Casting Yield: yours 65.0% vs industry 65–85%
Action 1	Work with die designer to optimise runner/gate geometry and reduce pour weight
Action 2	Consider vacuum-assisted HPDC (vacural) to allow thinner gates and reduce gating volume

[INFO] Material cost is below benchmark range

Finding	Material at 19.2% is below the typical 35–60% range. Conversion cost is the dominant driver.
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Impact	Low
Benchmark	Material %: yours 19.2% vs industry 35–60%
Action 1	Focus optimisation efforts on process and labour efficiency

§12 — Cost-Reduction Opportunities (ranked by saving)

7 opportunities · combined ¥27.01/part (~13%)

Ranked on money per part against this costing. The combined figure is the root-sum-square of the top three, capped at 40% — overlapping actions do not save twice, so the column is deliberately not summed.

#	Category	Action	Why it is on the list	Save/p art	%	Timeframe	Risk
1	Material	Improve Material Utilisation via Near-Net-Shape	Current utilisation at 65%. Near-net-shape process or improved nesting targets 80–90%.	¥20.62	10.0 %	Medium Term	Medium
2	Material	Consumables Rationalisation (Cores, Patterns, Shell)	Per-part consumables are 34% of the material line. Core count, pattern life and shell recipe are all design-controllable.	¥12.37	6.0%	Medium Term	Medium
3	Material	Scrap Revenue Recovery at Index Prices	35% of the bought material leaves as chips, offcuts or skeleton. That scrap has an index price — the credit belongs in the piece price, not in the supplier's margin.	¥6.70	3.3%	Quick Win	Low
4	Design & Geometry	Part-Count / Feature Integration Review (Boothroyd)	6 operations suggest features or joined parts that may not need to exist separately. The Boothroyd minimum-part-count test asks of each: does it move, must it be a different material, must it be separate for assembly?	¥12.37	6.0%	Long Term	Medium
5	Design & Geometry	Tolerance and Surface-Finish Relaxation	Machining is 36.3% of the part. The tightest tolerance and finish callouts set the machine, the cycle and the inspection burden — reviewing them is the cheapest cost lever that exists.	¥8.25	4.0%	Quick Win	Low
6	Process & Automation	Unmanned / Lights-Out Running	Labour is 16.0% of cost on a machine-paced part with healthy OEE (87%) — the classic profile for bar-feed or pallet-pool unattended running.	¥9.89	4.8%	Medium Term	Medium
7	Process & Automation	Multi-Machine Manning	Operations are manned at up to 1.0 operators per machine while the machine paces the cycle. One operator tending two machines is the textbook correction.	¥8.24	4.0%	Quick Win	Low

§13 — Opportunity by Category

3 categories with an actionable lever

Category	Opportunities	Best single action	Best save/part	Category total (indicative)
Material	3	Improve Material Utilisation via Near-Net-Shape	¥20.62	¥39.70
Design & Geometry	2	Part-Count / Feature Integration Review (Boothroyd)	¥12.37	¥20.62
Process & Automation	2	Unmanned / Lights-Out Running	¥9.89	¥18.13

Item	What the costing assumed	How to close it
Unusually low material content — verify alloy grade	Material at 19.2% of part cost is below the 35% floor for castings. Verify alloy grade and net weight inputs.	Validate material weight and alloy price inputs against purchase orders

§15 — Implementation Roadmap

Quick Wins — Implement Immediately

- Tolerance and Surface-Finish Relaxation
- Multi-Machine Manning
- Scrap Revenue Recovery at Index Prices

Long-Term Changes — Strategic Investment

- Part-Count / Feature Integration Review (Boothroyd)